

# Token Incentive and Blockchain Developer Contribution

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## ABSTRACT

The success of a Blockchain venture critically hinges on the contribution of blockchain developers. Blockchain ventures are typically developed as open-source projects, where some developers contribute voluntarily, while others are motivated by token rewards as an incentive to participate. However, it remains unclear whether tokens can serve as a sustainable and effective incentive in the current volatile cryptocurrency market, where opportunistic activities are prevalent. To address this question, we leverage a unique setting where token incentives are substantially affected during Pump-and-dump (P&D) events. We examine how developers react to this impact and its potential influence on the blockchain development community. Utilizing the most granular code-development level data from GitHub, we construct two metrics to gauge developer contribution: contribution quantity and originality. We estimate the effect of token incentive on contribution through the interactive fixed effects counterfactual (IFEct) model. The results imply that developers contribute less frequently with lower code originality to the projects due to token value distortion after a P&D event. Moreover, we identify the moderation effect of varying degrees of decentralization. The findings suggest that a decreased degree of decentralization can amplify the negative effect of P&D, leading to the “Tragedy of the Commons”, thereby further diminishing developers’ motivation to contribute. Our research provides important insights into the effectiveness of token incentives in blockchain ventures, underscoring the need for a more sustainable and effective incentive system.

**Keywords:** Token Incentive, Blockchain Venture, Developer Contribution, Pump and Dump, GitHub Code Analysis, Decentralization

# 1. INTRODUCTION

How to incentivize employees to contribute is a crucial challenge across various organizational contexts. Extensive research has attempted to identify effective and sustainable incentive methods to motivate individuals to contribute, both for formal companies (e.g., Kole, 1997) and informal organizations, such as an open-source community (e.g., Hann et al., 2002). For instance, public and private companies often use monetary incentives such as stock options to compensate executives and employees (Carey & Ward, 1999). In contrast, open-source communities, such as Linux and GitHub, where contributors are usually not formal employees bonded by labor contracts, rely more on non-monetary incentives such as badges, reputation, and name recognition (Lerner & Tirole, 2002).

Nonetheless, the emergence of blockchain spawns a new form of organization structure, blockchain venture<sup>1</sup>, that differs from conventional centralized corporations or open-source communities. Although most blockchain ventures (e.g., that of Bitcoin and Ethereum) rely heavily on the contribution of open-source developers, they cannot be strictly classified as either traditional open-source communities or conventional corporations due to their unique organization structure. Typically, these blockchain ventures have no hierarchical management structure and can even achieve fully decentralized governance, as exemplified by Bitcoin's use of a peer-to-peer network to facilitate decentralized network governance (DNG)<sup>2</sup>, which disperses authority across a network of participants in lieu of bureaucrats and centralization (Nakamoto, 2008; Zwitter & Hazenberg,

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<sup>1</sup> We use the term blockchain venture to refer to these blockchain-enabled enterprises that conduct their entire operations on blockchain and publish their codes on GitHub. These organizations are often listed on CoinMarketCap, CoinGecko, etc.

<sup>2</sup> We follow the literature and define DNG as the governance of a blockchain network of participants (who do not necessarily know each other) through a network in which governance task is distributed to a set of decentralized entities (Zwitter & Hazenberg, 2020).

2020). The scalable network governance in blockchain promotes a decentralized collaboration rather than a centralized government, attracting contributors rather than employees.

However, unlike contributors in a pure open-source community, contributions (e.g., code written by developers) in blockchain ventures are not always free; rather, many of them are rewarded with tokens (e.g., Ripple's bounty program). In fact, some contributors in blockchain ventures are token holders, e.g., Ethereum founder and core developer Vitalik Buterin owns more than \$1 billion worth of Ether (Cao, 2021); Basic Attention Token (BAT) explicitly sets aside 200 million tokens for its development pool (BAT, 2017). And many contributors participate in a blockchain project in pursuit of such token rewards, e.g., content contributors and curators at Steemit; the popular X-to-Earn model (Keller, 2022) allows users to earn cryptocurrency by engaging in meaningful activities. As a result, token rewards play a significant role in incentivizing contributors to participate in blockchain ventures.

Many even perceive blockchain ventures as a token-based incentive system, dubbed "token economy" (Kampakis, 2018). Tokens, e.g., those in the form of cryptocurrency, represent an internal property within a blockchain-enabled organization. They provide participants with a stake in the network and hence motivate them to contribute. They can be used to represent digital assets or access rights to reward certain activities or to engage in economic transactions inside or outside the organizations, such as asset transferring or developer hiring (Voshmgir, 2019). In addition, many tokens can be liquidated on cryptocurrency trading platforms such as Coinbase and Binance. In a sense, token holders are property owners via cryptocurrencies, distinguishing blockchain contributors from open-source community contributors. Blockchain contributors also differ from corporate employees since they are not constrained by labor contracts or managed by centralized

hierarchical institutions. Given the unique structure of blockchain ventures, how to incentivize individuals to contribute poses a new challenge.

At a broad level, this study seeks to investigate whether tokens are a viable incentive for individuals in blockchain ventures. However, how much incentive a token provides to individual contributors depends on the (potential) value of the token. Determining the value of a token (token valuation) presents a new challenge both in the academy and practice in the digital economy, as blockchain ventures often lack information (in contrast to a stock's fundamentals) to be accurately evaluated, leading to high volatility in cryptocurrency prices. The diverse opinions on token valuation among the public and the information asymmetry between blockchain founders, developers, and investors only exacerbate it. All these spawn arbitrage opportunities and, in many cases, even lead to the often-criticized pump-and-dump (P&D) schemes. Furthermore, the absence of labor contracts and explicit laws to regulate token distribution has led to suspicion of decentralization for some blockchain projects (e.g., when a significant proportion of tokens is held by a small number of holders). As a result, inappropriate token distribution may induce dishonesty and opportunistic self-interest behaviors, leading to the “Tragedy of the Commons”<sup>3</sup>, ultimately damaging the promising projects. Though token was initially invented to align self-interest with common welfare (Nakamoto, 2008), it may not serve as a sustainable and effective incentive.

Our first research question thus investigates **whether tokens serve as a reliable incentive for developer contribution**. To answer this, we employ a unique identification strategy. This strategy involves identifying a setting in which the token incentive is significantly impacted, and

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<sup>3</sup> The term “Tragedy of the Commons” was first used by Garrett Hardin in 1968 (Hardin, 1968). It describes a situation, in which individuals act independently based on their own self-interest and contrary to the common welfare of all users, can eventually destroy a shared resource and be detrimental to all individuals (Investopedia, 2022).

then observing how blockchain developers react to this impact. In the cryptocurrency market, the prevalent pump-and-dump (P&D) activities are such identifiable events that significantly distort the token price, providing an opportunity to examine the impact of P&D on blockchain developer contributions. Through this unique lens, our research offers a bird's-eye view of the potential flaws associated with using tokens as incentives. P&D is widespread (and notorious) in cryptocurrency trading. Hamrick et al. (2018) identified close to 5,000 P&D events that occurred on Discord and Telegram in 2018. P&D investors usually gather on Discord, WallStreetBets and Telegram groups and manipulate cryptocurrency prices by first pushing the price to a new peak, attracting more investors, and then dumping it within a short period of time (with the average P&D duration being 139 seconds). While a P&D event can increase the value of cryptocurrency (albeit shortly) and attract public attention, thus encouraging blockchain developers to contribute more, developers may choose to cash out immediately or even be disincentivized after witnessing the token being dumped. Consequently, they may reduce their contribution level or even quit the project under development. Moreover, P&D can harm the token's reputation, which may further discourage developers from continuing to contribute. Due to these counteracting forces, it remains to be seen how token incentive under the influence of P&D affects developer contributions and the overall technological progress of the blockchain venture.

Our second research question examines **whether the impact of token incentives on developer contribution varies on the level of decentralization in token distribution**. In a blockchain venture, the degree of decentralization is largely determined by the distribution of tokens among stakeholders. If tokens are held by only a few individuals, the level of decentralization is low, and vice versa. The effects of decentralization have been studied in traditional corporate structures,

where a high level of centralization is found to hinder innovation and demotivate employees, while decentralization can motivate and attract dedicated employees (Hamel, 2011). Accordingly, it stands to reason that within a blockchain venture context, token incentives may exert a heterogeneous effect on developer contribution when the level of decentralization varies.

Our empirical investigation uses data from four sources. *First*, we obtained all the 1,339 P&D events that occurred at Telegram for 403 cryptocurrencies from PumpOlymp between June 2018 and July 2021. *Second*, we acquired data on market capitalization, price, 24-hour trading volume, and aggregated developer contribution (e.g., number of commits) from CoinGecko for each cryptocurrency. *Third*, we retrieved GitHub data for all the existing blockchain ventures that have active development communities on GitHub. *Finally*, we collected token distribution data for each cryptocurrency listed on CoinMarketCap.

Our empirical analysis aims to **examine how developer contributions changed before and after token incentive distortion due to the exogenous P&D events compared to developers whose tokens did not experience distortion**. To measure developer contribution, we propose two novel metrics: **contribution quantity** and **originality**. Contribution quantity is measured by the number of developers' commitments to the blockchain ventures, while contribution originality is measured by the originality of source codes. To account for potential time-varying and unit-varying confounders, we perform **coarsened exact matching (CEM)** to ensure that the treatment and control groups have comparable baseline characteristics. We then apply the *interactive fixed effects counterfactual* (IFEct) model to estimate the effect of P&D-induced incentive distortion on developer contribution. Further, we examine the heterogeneous effects arising from the varying degrees of decentralization.

The IFect results reveal that developers contribute less frequently and with lower originality to the projects after their token incentive is distorted by P&D. This effect is moderated by the degree of decentralization: a decrease in the degree of decentralization (increase in the degree of centralization) exacerbates the negative impact of P&D, resulting in a further decrease in developer contribution. A low degree of decentralization contradicts the fundamental principle of blockchain. As a result, opportunistic events like P&D can further induce individual dishonesty, distrust, and self-interest acts, thereby reducing developers' motivation to contribute, leading to the "Tragedy of the Commons".

Our research contributes to the extant incentive literature in industrial organization by identifying potential flaws in the token-economy based incentive system. Under the present cryptocurrency market, where opportunistic events are widespread, tokens may not serve as a sustainable incentive to maintain the healthy growth of the blockchain community. Furthermore, we enrich the literature on decentralization by demonstrating how a low degree of decentralization can undermine the incentives for blockchain developers and hinder technological innovation, differing from the prior work (Benedetti & Kostovetsky, 2021; Lee et al., 2022; Lyandres et al., 2022). Lastly, we make a contribution to the token valuation literature by providing strategies to evaluate blockchain projects. Our evaluation of source code production provides a viable way to quantify developer contribution and the value of a blockchain venture.

## **2. LITERATURE REVIEW AND THEORETICAL FOUNDATION**

### **2.1. Financial and non-Financial Incentives**

The literature on incentives has extensively studied the role of both monetary and non-monetary incentives in corporations and open-source communities. For corporations, stock options have

been a common monetary incentive to motivate employees and increase firm value (Core & Guay, 2001). These options typically have a vesting period of several years to elongate their effectiveness over time (Kole, 1997). Empirical evidence from Cowgill and Zitzewitz (2009) suggests that well-designed stock option plans can increase employee retention. However, the study of Oyer and Schaefer (2005) provides contradictory evidence, suggesting that stock options may be an ineffective incentive unless employees are optimistic about the company's future.

In the context of open-source communities, Malgonde et al. (2022) find that during the COVID-19 pandemic, developers tend to contribute more to others' projects than to their own; yet, developers tend to contribute less to others' projects when facing the threat of rising unemployment. Stein et al. (2022) suggest that online communities may lack sustainability in the long run due to the challenge of aligning individuals and collective interests through a longitudinal case study. Chen et al. (2022) explore the impact of technological acquisitions on firm-sponsored open-source software (OSS) projects, revealing that external contributors increase their activity post-acquisition, while internal contributors shift focus to other OSS projects. Lerner and Tirole (2002) argue that a programmer will choose to contribute only if the benefits outweigh the costs. But these benefits may not necessarily be monetary and can include immediate gratification from achievements and delayed benefits such as name recognition or job opportunities in the future. Burtch et al. (2022) suggest that peer awards motivate user to generate more novel content on online communities. Nevertheless, the findings are inconclusive in the literature. Hann et al. (2002) find that greater open-source participation does not always lead to tangible wage increases. Some studies suggest that monetary incentives are still important for open-source content generation



(Wang et al., 2022), whereas other studies (Qiao et al., 2021; Sun et al., 2017) find the opposite or no effect of using monetary incentives.

The use of blockchain technology has spawned a new form of organization structure that differs from traditional hierarchical corporations or open-source communities. The above stream of literature has not yet studied how developers are incentivized in a blockchain venture. We add to this literature by exploring the role of token incentives on blockchain developers.

## **2.2. Token Incentive in Token Economics**

Incentive design is central to all blockchain ventures that rely on user contributions to succeed (Barrera, 2019; Hurder, 2020). Many of these ventures use tokens as monetary incentives to reward users and steer them to contribute. Such a token-based mechanism gives rise to an emerging field called *token economics* that focuses on the design of a token mechanism to incentivize users to act towards a common goal of a blockchain platform (Voshmgir, 2019). For example, one of the desired goals is to distribute tokens to users equitably based on the quality and quantity of their contributions. The bitcoin system, for instance, distributes bitcoins to miners according to their contributions to solving a cryptographic puzzle, effectively steering them toward the common goal of maintaining its distributed ledger. In token economics, how to align token incentives with user contributions and regulate user behavior is critical to building a successful blockchain venture (Chen et al., 2021; Furlonger & Uzureau, 2019). The difficulty arises, however, from the simple fact that users are self-interested, and their actions may not always be aligned with the common goal of the platform. Governance without a central authority (i.e., decentralized governance) only exacerbates such tensions.

Tokens play a vital role in a blockchain project, as they help maintain the sustainable growth of the venture and incentivize desirable behaviors of participants (Kampakis, 2018). Take Bitcoin as an example. The introduction of bitcoin cryptocurrency motivates participants to maintain, manage and expand the distributed ledger (Nakamoto, 2008). The use of tokens enables the creation of a “governance market” where individuals can buy and sell governance through tokens and are awarded tokens for their contributions to decentralize governance at scale. Take the X-to-Earn model as an example. Individuals are rewarded with cryptos to perform certain X activities, such as play-to-earn. As the user network expands, the value of the token increases, and the benefits distributed to the participating token-holders increase (Kim & Chung, 2018).

Previous studies have mainly focused on studying the design of different types of tokens, such as utility tokens, equity tokens, and security tokens (Conley, 2017; Kampakis, 2018; Kim & Chung, 2018), the regulation of tokens in a decentralized organization (Ferreira, 2020), enhancing risk management through blockchain technology (Zheng et al., 2022), promoting airdrop in initial coin offerings (Li et al., 2021), and the revolutionary impact of blockchain through tokens (Lee, 2019; Sockin & Xiong, 2023).

However, there lacks empirical evidence to validate the effectiveness of using tokens as incentives, especially during the cryptocurrency frenzy in which opportunistic events such as P&D are prevalent. Our study fills this vacancy by demonstrating the potential flaws in the current token incentive system that may hinder continued developer contribution and the healthy growth of the blockchain development community. Additionally, our study provides new means for cryptocurrency valuation by analyzing the source code contributed by blockchain developers. Ultimately, the value of a blockchain venture depends on the collective contribution of the

developers. This is in sharp contrast to the current cryptocurrency valuation literature, which mainly focuses on the financial values (e.g., market valuation) of tokens (Cong et al., 2021; Conley, 2017; Shen et al., 2020).

### **2.3. Impact of Incentive Distortion on Developers**

Another relevant stream of studies is price manipulation. In the stock market price manipulation literature, scholars primarily focus on the impact of P&D on market performance. Huang and Cheng (2015) illustrate that P&D manipulations lead to high temporary price impacts, increased volatility, larger trading volumes, and higher market inefficiency. Bouraoui (2015) examines a sample of penny stocks and finds that stock price manipulation can generate a significant increase in returns on the first day, followed by a significant decrease during the following days.

In the cryptocurrency market price manipulation literature, several studies have developed algorithms to detect price manipulation. Mirtaheri et al. (2021) utilize social media data from Twitter and Telegram to detect P&D schemes. Chen et al. (2019) modify the Apriori algorithm to identify abnormal trading behaviors and user groups that are indicative of a P&D. Kamps and Kleinberg (2018) employ trading signals to detect P&D schemes. Xu and Livshits (2019) use a random forest model to predict the likelihood of a coin being pumped before the P&D event occurs. Some other studies investigate the impact of P&D on the cryptocurrency market. Li et al. (2021) suggest that P&D leads to a short-term increase in prices, volume, and volatility. Hamrick et al. (2018) find that market capitalization and volume are the most important factors in predicting the returns of P&D events.

What is notably missing in the literature is an investigation of the impact of price manipulation on developers' incentives to contribute to a blockchain project. Manipulation like

P&D distorts the true value of a token and alters developers' expectations regarding the token incentive, hence affecting their willingness to contribute to code development. Our study fills this literature gap.

Identifying the impact of token incentives on developer contributions requires exogenous variation on token incentives. We leverage the shock of P&D events for this purpose. P&D changes the token incentive for developers, which in turn affects their contributions. A P&D event would push the cryptocurrency price to a new peak, and developers are likely to use this recent peak price as a new anchoring point to form their expectation of the token reward. As a result, developers tend to have higher expectations of the token, having seen a higher peak after a P&D. However, P&D events often lead to a short period of price peak followed by a significant decrease in a short period of time (Bouraoui, 2015). The sharp fall of the cryptocurrency value following a P&D (downwardly) negatively violates developers' expectations. According to *expectancy violation theory*, negative violations of expectations are perceived more negatively than expected (Burgoon, 2015; Burgoon & Jones, 1976). Violations tend to stand out and are remembered because people pay more attention to them (Lannutti et al., 2001). Therefore, people's motivation can be impaired by expectation violations (Furlich, 2016). We expect that developers' decreased expectation of token value due to P&D events will lower their motivation to contribute. This is exemplified by the case of CloakCoin, which was once considered one of the most promising projects with a \$6 million market cap in July 2014. Its price was pushed up to a new peak during P&D and then plunged (Figure F3 in Appendix F). When P&D was over, the original development team was unable to cope with the disappointment and finally left the project in October 2014 (Scott, 2015).

## **2.4. Moderating Effect of Degree of Decentralization**

The governance in blockchain applications fundamentally differs from that of traditional platforms under the “platform economy” where a centralized authority, such as platform owners or managers, exerts controlling power on user activities (Constantinides & Barrett, 2015; Huber et al., 2017). In contrast, blockchain ventures typically embrace decentralized governance through tokenization to resolve conflicts between platform and users (Sockin & Xiong, 2023), preventing user exploitation by delegating governing powers from the platform owners to users, e.g., by distributing tokens to contributing users in the case of Steemit’s Proof-of-Brain mechanism or allocating voting rights to nodes in the case of EOS’s Delegated Proof of Stake system. There is a long-standing debate on the pros and cons of centralized versus decentralized structures in organizations. Previous studies in management (Leavitt & Kaufman, 2003; Pfeffer, 2013) indicate that hierarchy forms the foundation of most organizations, not just because of its efficiency, but also because people genuinely favor and embrace hierarchy. However, a high level of centralization can impede innovation and spawn distrust in organizations (Hamel, 2011). In contrast, a decentralized organization structure may help empower and motivate employees, unleash their creativity and improve their loyalty to the organization (Hamel, 2011; Malone, 1997).

Recently the trend of decentralizing organizations has gained momentum as blockchain continues to penetrate businesses. Blockchain empowers a new type of organization where the conventional top-down hierarchical governance is replaced with a bottom-up *DNG* (De Filippi, 2017; Zwitter & Hazenberg, 2020), by leveraging the resources (e.g., computing power) contributed by a peer-to-peer network consisting of individual nodes. These nodes collaborate and contribute their resources to the organization through consensus mechanisms, for which tokens are

commonly used to incentivize these nodes to contribute and enforce decentralized governance at scale (Nakamoto, 2008). Thus, determining how to distribute tokens to entities under the consensus mechanism is critical to ensure that these self-interested individuals can collaborate toward a common goal without the need to trust each other (Economist, 2015; Maples, 2018). The degree of decentralization manifests through the distribution of the tokens. An organization where a large portion of tokens are owned by a small number of holders tends to have a lower degree of decentralization, i.e., lower level of *DNG*, and vice versa.

In a typical blockchain venture, participants are not bonded by labor contracts. Therefore, the number and value of tokens they receive serve as the primary incentive for them to contribute. However, there are currently no regulations or laws to restrict how tokens should be distributed. In practice, token distribution is primarily determined during the (unregulated) ICO. For example, Younow.com, the first ICO legally approved by the SEC, specified the distribution of its Props token as follows: founders reserve 26% of the tokens, public investors hold 20%, advisors receive 4%, ecosystem infrastructure takes 10% and contributors (including code developers) receive the lion's share of 40%. Clearly, the percentage of tokens to be awarded to contributions is critical for incentivizing them to contribute. A highly concentrated token distribution (i.e., the majority of tokens owned by a few people) may raise several concerns for contributors: 1) it leads to centralized ownership, which goes against the core principle of blockchain ventures; 2) such tokens are more likely to be subject to fraud (Grobys, 2020); 3) it may induce self-interest behaviors and distrust among participants, leading to the "Tragedy of the Commons" (Maples, 2018) that may inflict *DNG*. Platforms that depend on shared and unregulated resources will

ultimately suffer if individuals are incentivized to pursue self-interest instead of achieving the common goal of the community.

Specifically, developers in a blockchain venture with a low degree of decentralization would have less control over the development of the blockchain project since they have a smaller token holding relative to other stakeholders. Developers would have less trust in such projects. As a result, when their incentives, tokens, are distorted by exogenous opportunistic events such as P&D, they are more likely to cast doubt on the future of the venture and scale down or even discontinue their contributions to the project. Therefore, we conjecture that the impact of P&D on developer contribution would be moderated by the varying degree of decentralization.

### **3. DATA AND VARIABLES**

#### **3.1. Data**

Our data come from four sources: PumpOlymp, CoinGecko, GitHub, and CoinMarketCap.

##### **PumpOlymp Data**

PumpOlymp is a platform that tracks historical P&D events. We obtained all the existing 1,339 P&D events that occurred at Telegram for 403 tokens from June 2018 to July 2021. The data track event date, Telegram channel ID, exchange platform, duration, volume, the price before the P&D, the max price during the P&D, theoretical profit, and token symbol. A token symbol is similar to a ticker symbol in the stock market, uniquely identifying the token. Table C1 in Appendix C provides the summary statistics for all the events. On average, a P&D event lasts about three minutes, and more than 90% of them succeed with positive returns.

### **CoinGecko Data**

CoinGecko tracks cryptocurrencies' trading transactions and development activities from GitHub. We collected trading data, including market capitalization, price, and 24-hour volume. The development data contains the number of commits, total issues, and subscribers for all the existing 5,990 cryptocurrencies from April 2013 to December 2020. Table C2 presents the summary statistics of the relevant variables.

### **GitHub Data**

There were 2,351 blockchain ventures (token projects) on GitHub when we retrieved the data in April 2021. The GitHub data includes the publish date of each project, the version and date of each update, the original source code, and the updated source code for each version, spanning from 2011 to April 2021. The GitHub source codes for these blockchain ventures cover 49 different programming languages, such as C++, C, Python, JavaScript, Solidity, etc. We focus on the commonly used programming languages, including C++, C, Python, JavaScript, TypeScript, Go, Solidity, and Java, which account for more than 75% of blockchain ventures. In total, our codebase data contains more than 1.27 billion lines of code and 18,010 code files (Figure F1 in Appendix F). Table C3 in Appendix C shows the distribution of blockchain ventures using different programming languages.

### **CoinMarketCap Data**

CoinMarketCap is a platform that aggregates cryptocurrencies' trading information from various exchange platforms. CoinMarketCap collects data such as market capitalization, price, and, most importantly, holder statistics for each cryptocurrency. The holder statistics data include the percentage of tokens held by the top 10 holders, top 20 holders, top 50 holders, and top 100 holders.



We collected all the existing holder statistics listed on CoinMarketCap. Table C4 in Appendix C shows the summary of holder statistics for all the 3,555 tokens we collect.

Since our data come from multiple platforms, we use cryptocurrencies' trading symbols and token IDs for matching. After cross-platform matching, we end up with 811 P&D events for 255 cryptocurrencies, which form our treatment group. For the control group, we exclude these nonsignificant tokens with a market capitalization of less than \$1,000. As a result, we are left with 2,309 control cryptocurrencies.

### **3.2. Data Matching**

We then pre-processed the data to match the control and treatment units using the Coarsened Exact Matching (CEM). This is to ensure that the baseline characteristics of the two groups are closely matched, so as to alleviate the endogeneity concern that the control and treatment units might be systematically different due to organizers' selection of certain tokens to pump and dump. In terms of matching criteria, previous literature in P&D suggests that P&D organizers might consider market capitalization, trading volume, and volatility when choosing a target (Kamps & Kleinberg, 2018; Li et al., 2021). Thus, we select four trade-level criteria: market capitalization, trading volume, volatility, and weekly return for matching. In particular, we choose one month and one week prior to the P&D event, i.e., [-37, -8], and leave out the final week as our matching period, following the previous study (Li et al., 2021). The final week is excluded because P&D organizers may pre-load the target coin one week [-7, -1] before the event, which may induce abnormal trading activity (Li et al., 2021).

We conduct two CEM matchings, one for contribution quantity and another for contribution originality, respectively <sup>4</sup>. For contribution quantity, among the 255 treated cryptocurrencies, 90 tokens either do not have an open-source development community or lack commitment records. Meanwhile, the IFect model requires at least 5 pre-treatment periods. These two constraints leave us with 148 treated tokens. Some of the 2,309 control tokens have not yet started public trading on token exchanges (e.g., Coinbase) during the sample period and thus are removed. Accordingly, we are left with 900 control tokens. We apply the CEM matching to these 148 and 900 tokens in our treatment and control groups. For each trade-level matching variable, the coarsened optimal number of bins is determined by the Sturges formula. After matching, our treatment group consists of 123 treated units, whereas the control group comprises 260 control units. Table C5 in Appendix C summarizes the covariate balance for each variable before and after matching. After matching, our covariate balance for each variable has improved by more than 90%.

For contribution originality, focusing on the eight programming languages as elaborated in the next section and the constraint on the IFect model, which requires at least 5 pre-treatment periods, left us with 104 treated and 864 control tokens. After CEM matching, 100 treated and 244 control tokens remain. Table C6 in Appendix C provides the covariate balance before and after matching, showing that the covariate balance for each variable has improved by more than 90%.

### **3.3. Dependent Variable: Developer Contribution Quantity and Originality**

We construct our dependent variable by investigating developer contribution to the GitHub code of a blockchain venture. Specifically, we propose two metrics that capture contribution quantity

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<sup>4</sup> We conduct two matchings because the constraint on eight programming languages has left us with different numbers of treated and control tokens for contribution quantity and originality. Furthermore, our contribution quantity and contribution originality are derived from two different features on GitHub. We will elaborate on how we derive these two measures in Section 3.3 in detail.

and originality. These two measures are derived from two different features tracked on GitHub: the number of commits and version update per release. Appendix B provides a detailed explanation of these two GitHub features, as well as why and how we constructed these two measures using these two different features.

Contribution quantity is measured as the number of GitHub commitments (called commits) contributed by developers to a token in a week. We note that prior research on GitHub has often used the number of commitments to indicate source code quality (Fisch, 2019; Petryk et al., 2022; Ray et al., 2014). Some recent studies on ICO simply use the number of commitments to approximate blockchain project quality (Davydiuk et al., 2023; Lyandres et al., 2022). However, we maintain that it is questionable to use this quantity measure to signify a blockchain project’s quality, especially when the similarity between source codes is high. According to Lopes et al. (2017), over 70% of GitHub codes are duplicates of earlier files.

We propose a new metric, originality, that captures the quality of developer contribution. For each GitHub version update, we compare its source code to other pre-existing source code files and quantify their similarity. The originality score is defined as the opposite of the similarity score. This originality measurement is inspired by the computer science literature on GitHub clone detection and source code analysis (Lopes et al., 2017; Svajlenko & Roy, 2015). Lopes et al. (2017) argue that studies on software quality using commit tags, such as “bug(s)” and “defects”, to determine project quality fail to account for file-level code duplication and, thus, can be poor-quality measures. A quality indicator that does not address code similarity concerns cannot accurately reflect the quality of a project. Code duplication can be a serious problem for blockchain ventures due to their non-copyrighted, open-sourced nature of code. For example, the popular

meme coin, Dogecoin, was created in December 2013 and was essentially a copy of Bitcoin (Schroeder, 2022) rather than a genuine technology breakthrough.

Our originality score directly accounts for code similarity by comparing the updated source code of a token in a month to pre-existing source codes of other tokens<sup>5</sup>. The comparison is done using *Moss* (Measure of Software Similarity), an authoritative code similarity detection software developed by Stanford University. Moss has been widely used by numerous academic and industry institutions for over a decade, especially in copyright detection, for its broad coverage of programming languages (28 programming languages currently). For each version update in each month, we use Moss to generate a code similarity score measured based on the percentage of source code that is similar to other source codes. Our definition of originality score is the opposite of source code similarity score (i.e.,  $100\% - \text{similarity score}$ ). A higher originality score indicates higher quality. We focus on the top eight programming languages used by blockchain ventures, including C++, GO, JavaScript, TypeScript, C, Java, Python, and Solidity, which cover over 75% of the current blockchain ventures on GitHub (See Table C3 in Appendix C). In total, our code base includes more than 1.27 billion lines of code and 18,010 code files.

Table C7 in Appendix C summarizes the mean difference for dependent variables between treatment and control groups before and after the P&D. Simply inferring from the difference in means, contribution quantity has decreased around 21.58 on average after token distortion, while contribution originality has decreased around 14.18 on average.

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<sup>5</sup> Please refer to Appendix B for an explanation on why it is necessary to use different level of granularity for the two measures: weekly for contribution quantity while monthly for contribution originality.

### 3.4. Independent and Control Variables

Our main independent variable is a dummy variable indicating whether, for a token, the P&D occurred in a period (week or month). Some may argue that P&D may not be exogenous, and P&D organizers may specifically target certain tokens (e.g., those with low market capitalization or high return). However, this is unlikely to be the case, as suggested by both the literature and anecdotal evidence. Hamrick et al. (2018) identified around 5,000 P&D events on Telegram and Discord and found that organizers **tend to use false news or rumors to promote coins, rather than anything related to the feature of a token**. Li et al. (2019) and Tsuchiya (2021) suggest that P&D in the cryptocurrency market is distinct from traditional information-based P&D in the stock market, in that the positivity or negativity of the information determines how the P&D is carried out. In contrast, a cryptocurrency P&D often focuses on trade-based manipulation by increasing trade volume through inviting many participants, irrelevant of the nature of a token. That is, P&D organizers choose cryptocurrency rather randomly, instead of selectively based on token quality or market performance (see e.g., the YoBit example in Appendix A). We conduct a robustness check in Section 5.2 to rule out possible selection factors. Further, Appendix A provides more detailed discussions on the phenomenon of P&D in the cryptocurrency market with explanations on why P&D organizers **randomly choose tokens**.

Our selection of control variables is inspired by the cryptocurrency **three-factor model** (Liu et al., 2022), which includes **market capitalization, transaction volume, token price, and token return**. Market capitalization and transaction volume are used to control the size and weight of blockchain ventures, avoiding any possible impact caused by their market size. Token price is used to control potential market trends that could influence our dependent variables and to assess the

possible impact on developers caused by high or low token prices. Lastly, we include token return as a control variable to capture the potential impact on developers due to the amount of return they received. Table C8 and C9 in Appendix C present descriptive statistics for our control and dependent variables for contribution quantity at a weekly level and contribution originality at a monthly level.

## 4. EMPIRICAL ANALYSES AND MAIN RESULTS

Our empirical identification strategy involves examining the change in developer contributions, both in terms of quantity and originality, in reaction to P&D, which changes the token incentive perceived by developers. This creates a staggered difference-in-differences setting (DID) since P&D events may occur at different times for different tokens, and some tokens may experience multiple P&D events. Recently the validity of the staggered DID estimation has been questioned for its reliance on strict (or unrealistic) assumptions, including the pre-treatment parallel-trend assumption, no anticipatory assumption, and the homogenous treatment assumption (Baker et al., 2022). An alternative method that directly accounts for these assumptions is the Fixed Effects Counterfactual (FEct) method (Liu et al., 2022; Xu, 2017), which has been shown to outperform the two-way fixed effects (TWFE) and the two-stage DID estimators when these assumptions may not strictly hold (Borusyak et al., 2021; Gardner, 2022; Liu et al., 2022). Hence FEct is adopted in our empirical analysis <sup>6</sup>.

### 4.1. Econometrics Estimation Model

The specific estimation approach employed here is the Interactive Fixed Effects Counterfactual (IFEct), a variant of FEct that integrates the Synthetic Control model (Abadie et al., 2010) with

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<sup>6</sup> In addition to the IFEct model, we apply the difference-in-differences (DID) model specification with linear time trend in Section 5.3.

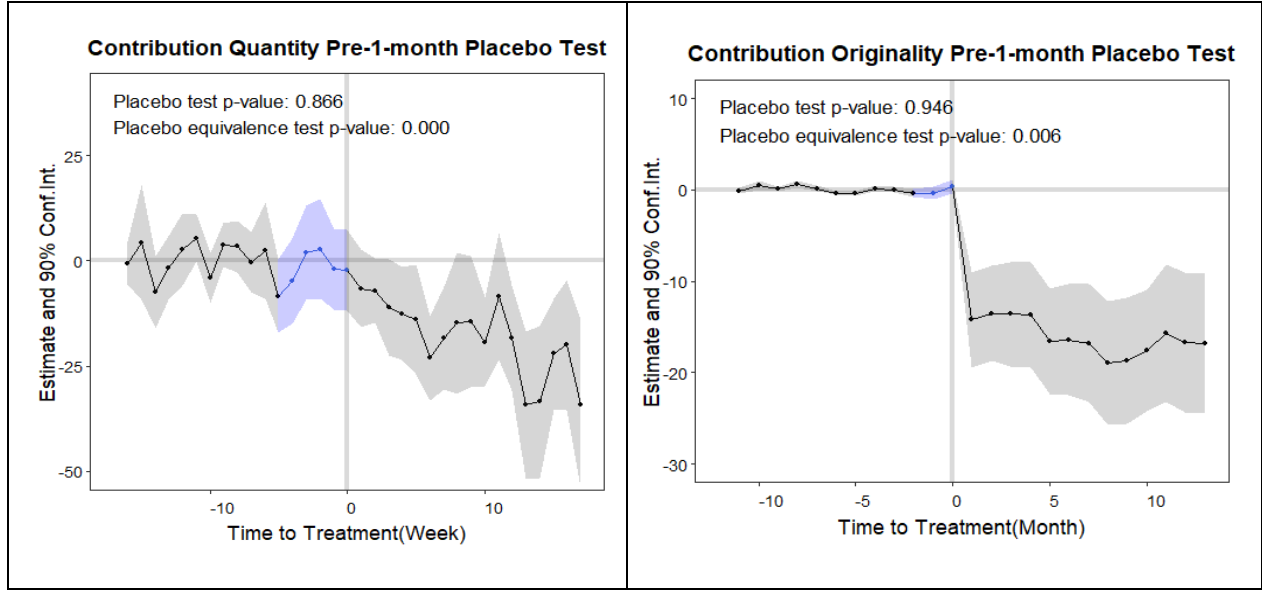
the Interactive Fixed Effects (IFE) model (Bai, 2009). IFECT estimator mitigates endogeneity arising from treatment heterogeneity by accounting for both the potential unobserved time-varying and unit-varying confounders. The model framework of IFECT can be expressed as Equation (1).

$$Y_{it} = \delta_{it}PD_{it} + x'_{it}\beta + \lambda'_if_t + \alpha_i + \xi_t + \varepsilon_{it} \quad (1)$$

$Y_{it}$  represents developers' contribution quantity and originality, measured by developers' weekly commitments and monthly source code originality, respectively, for token  $i$  at time  $t$ .  $PD_{it}$  indexes whether a token is experiencing a P&D at  $t$ .  $\delta_{it}$  is the heterogeneous treatment effect on unit  $i$  at time  $t$ .  $x'_{it}$  is a  $(k \times 1)$  vector of control variables as specified in Section 3.4, and  $\beta$  is a vector of corresponding parameters.  $\lambda_i = [\lambda_{i1}, \dots, \lambda_{ir}]'$  is an  $(r \times 1)$  vector of factor loadings while  $f_t$  is an  $(r \times 1)$  vector of unobserved common factors. The factor component takes a linear form and can be expressed as  $\lambda'_if_t = \lambda_{i1}f_{1t} + \lambda_{i2}f_{2t} + \dots + \lambda_{ir}f_{rt}$  to account for unobserved confounding factors. We add token-level fixed effects  $\alpha_i$  to account for time-invariant variations and  $\xi_t$  is the time fixed effects that captures any common time-specific shocks. The last term,  $\varepsilon_{it}$  is the unobserved error term.

Liu et al. (2022) developed several diagnostic tests to ensure the validity of the IFECT model, mainly including the Placebo and Equivalence tests. We examine the Placebo test considering two different placebo treatments for contribution quantity: 1) hypothetically assuming the treatment occurred 1 month prior to the actual treatment, and 2) assuming the treatment occurred 2 months earlier. For contribution originality, we consider two hypothetical scenarios when implementing the Placebo test by assuming the treatment occurred 1 or 3 months prior to the actual event, respectively. Table 1 columns (3)-(4) and Table 2 columns (3)-(4) present our estimations of the placebo effects. The magnitude of the placebo effects for contribution quantity

and originality all turns out to be close to 0 and statistically insignificant, passing the Placebo test. The results of the equivalence test in which the null hypothesis is reversed are depicted in Figure 1, which shows that the equivalence tests are passed.<sup>7</sup>

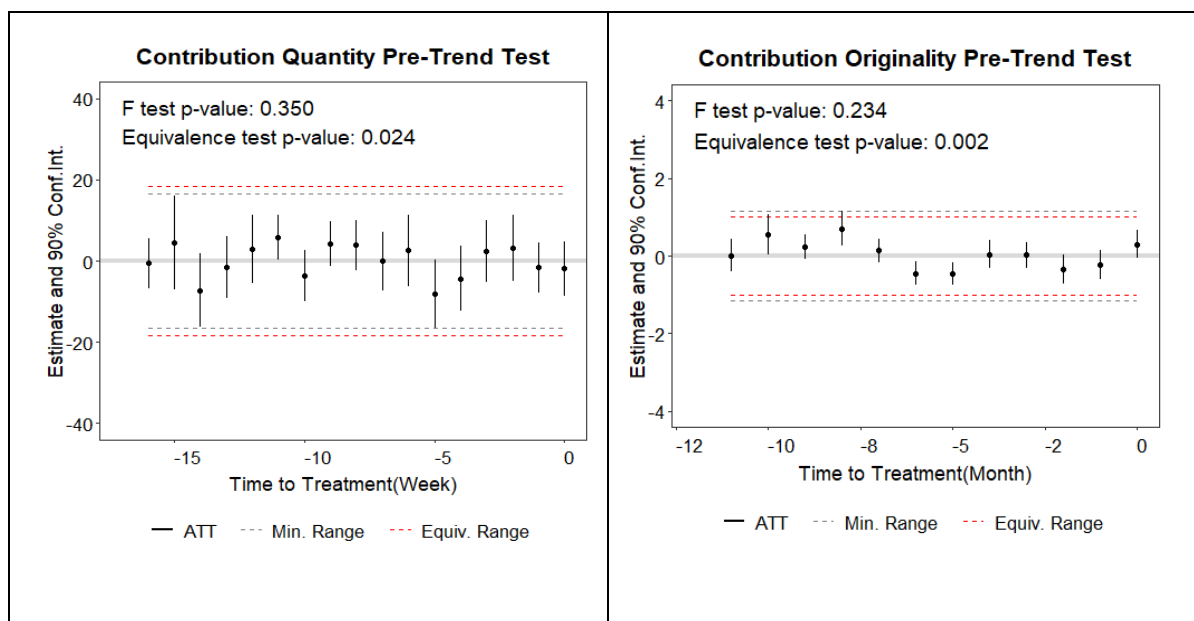


**Figure 1: Placebo & Equivalence Test for Contribution Quantity & Originality**

Additionally, we extend the pre-treatment period covered in the placebo test to the entire pre-treatment period to ensure parallel trend by accounting for the presence of possible time-varying factors that are uncontrolled. The results shown in Figure 2 indicate that we pass the no pre-trend and equivalence tests, which support the parallel trend assumption during the entire pre-treatment period and confirm that there is no anticipatory behavior prior to the treatment.

<sup>7</sup> The equivalence tests for other periods are reported in Appendix F, see Figure F2. Following Liu et al (2022) and Hartman and Hidalgo (2018), we set the equivalence range as  $[-0.36\hat{\sigma}_\varepsilon, 0.36\hat{\sigma}_\varepsilon]$ , where  $\hat{\sigma}_\varepsilon$  is the standard deviation of untreated residuals.





**Figure 2: No Pre-Trend & Equivalence Test for Contribution Quantity & Originality**

## 4.2. Impact on Developers' Contribution Quantity

We apply the IFect model to our CEM matched sample which consists of 123 treated tokens, and 260 control tokens. Table C8 in Appendix C summarizes the descriptive statistics of all the 383 tokens.

Table 1 (see Columns 1 & 2) reports our estimation of the effect of token incentive on developers' contribution quantity. Column 2 shows the results with time-varying variables: weekly average market capitalization, weekly average trading volume, weekly average token price, and weekly return as the controls. The main coefficient of interest pertaining to variable P&D, as shown in Column 2, is negative and statistically significant ( $P < 0.01$ ). Developers, whose incentive is distorted by P&D, contribute 18.29 fewer commits per week compared to their contribution level before the event, representing a 65.6% decrease on an average weekly commits of 27.86. This finding presents evidence that tokens may not serve as a reliable

incentive. The distortion of token price can demotivate developers, which can be detrimental to the development community in terms of contribution quantity.

**Table 1: Summary of Contribution Quantity Results**

|                                   | Contribution Quantity |                     |                  |                  |
|-----------------------------------|-----------------------|---------------------|------------------|------------------|
|                                   | (1)                   | (2)                 | (3)              | (4)              |
| P&D                               | -17.93***<br>(4.30)   | -18.29***<br>(4.36) |                  |                  |
| Placebo Pump 1-Month              |                       |                     | -0.853<br>(4.72) |                  |
| Placebo Pump 2-Month              |                       |                     |                  | -0.157<br>(4.22) |
| Weekly Avg. Price                 |                       | 0.0021<br>(0.074)   |                  |                  |
| Weekly Return                     |                       | -0.432<br>(0.29)    |                  |                  |
| Weekly Avg.MarketCap <sup>1</sup> |                       | -0.0007<br>(0.0036) |                  |                  |
| Weekly Avg.Volume <sup>1</sup>    |                       | 0.0035<br>(0.0076)  |                  |                  |
| Individual Fixed Effects          | Yes                   | Yes                 | Yes              | Yes              |
| Week Fixed Effects                | Yes                   | Yes                 | Yes              | Yes              |
| Observations                      | 13385                 | 13385               | 13385            | 13385            |
| Token Units                       | 383                   | 383                 | 383              | 383              |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . <sup>1</sup>The unit of market cap and volume is in million USD.

### 4.3. Impact on Developers' Contribution Originality

Similarly, for contribution originality, we apply the IFect model to the CEM-matched 344 tokens, which consist of 100 treated tokens, and 244 control tokens. Table C9 in Appendix C summarizes the descriptive statistics of these 344 tokens.

Table 2 reports our estimation of the effect on contribution originality. Column 2 shows the result estimated with the control variables. The negative and statistically significant coefficient of variable P&D (coeff. = -18.03,  $P < 0.01$ ) indicates that when token incentive is distorted by P&D,

developers contribute fewer original contents per month than they do prior to the event. On average, the originality score decreased by 18.03, a 25.8% percent decrease over an average score of 69.81.

**Table 2: Summary of Contribution Originality Results**

| Contribution Originality           |                     |                     |                   |                   |
|------------------------------------|---------------------|---------------------|-------------------|-------------------|
|                                    | (1)                 | (2)                 | (3)               | (4)               |
| P&D                                | -18.17***<br>(3.34) | -18.03***<br>(3.45) |                   |                   |
| Placebo Pump 1-Month               |                     |                     | 0.0263<br>(0.385) |                   |
| Placebo Pump 3-Month               |                     |                     |                   | -0.350<br>(0.424) |
| Monthly Avg.Price                  |                     | -0.0052<br>(0.0052) |                   |                   |
| Monthly Return                     |                     | -0.0015<br>(0.020)  |                   |                   |
| Monthly Avg.MarketCap <sup>1</sup> |                     | 0.0003<br>(0.0003)  |                   |                   |
| Monthly Avg.Volume <sup>1</sup>    |                     | -0.0001<br>(0.0007) |                   |                   |
| Individual Fixed Effects           | Yes                 | Yes                 | Yes               | Yes               |
| Month Fixed Effects                | Yes                 | Yes                 | Yes               | Yes               |
| Observations                       | 7221                | 7221                | 7221              | 7221              |
| Token Units                        | 344                 | 344                 | 344               | 344               |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . <sup>1</sup>The unit of market cap and volume is in million USD.

Together with our results on contribution quantity, our findings suggest that tokens may not serve as a sustainable incentive. The lack of fundamentals to stabilize the token value and the high fluctuation of token prices due to speculation, such as P&D, may mislead developers into having unrealistic expectations of the token. When their expectations are not met, developers may be demotivated to continue contributing. Blockchain ventures, thus, should be cautious when relying on tokens as the incentive to motivate developers, as monetary rewards like tokens may

lure opportunistic behaviors from developers and eventually hamper the technological innovation of the blockchain venture.

Our findings are echoed by Cointelegraph's interview with Cloak coin's core developer, Cashmen. Cashmen described how he started investing in Cloakcoin in July 2014 because of its high potential as one of the few promising blockchain projects reaching a \$6 million market cap. Then P&D haunted Cloackcoin in August 2014 (see Figure F3 in Appendix F), leading to an exodus of the original development team soon after (Scott, 2015).

#### 4.4. Heterogeneous Effect of Decentralization

We then examine the moderation effect of decentralization, particularly, how token distribution influences the contribution behaviors of developers. We derive the token distribution data from the top 10 holder statistics data<sup>8</sup>. The percentage of tokens owned by the top 10 holders represents the decentralization level in a blockchain organization. We stratify our data into two groups based on their token distribution: the top quartile and the bottom quartile. The top quartile consists of tokens ranked within the top 25% in terms of the degree of centralization (i.e., with a low degree of decentralization), while the bottom quartile is composed of tokens with the lowest 25% degree of centralization (i.e., with a high degree of decentralization). The middle quartile groups are dropped to ensure a sharper contrast in token distribution between the high and low groups. Equation (2) specifies the model with the interaction term added to account for the moderation effect of the degree of decentralization.

$$Y_{it} = \delta_{it}PD_{it} + \theta_{it}PD_{it} \times DD_i + x'_{it}\beta + \alpha_i + \xi_t + DD_i \times t + \varepsilon_{it} \quad (2)$$

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<sup>8</sup> We conduct several additional robustness checks using measure derived from top 20, top 50, and top 100 holder statistics. The results remain the same (available upon request).

$DD_i$  is a dummy variable indicating token  $i$ 's degree of decentralization, and it takes a value of 1 if token  $i$  belongs to the high centralization group and 0 otherwise. The term  $x'_{it}\beta$  captures the control variables as specified in Equation (1).  $\alpha_i$  and  $\xi_t$  are the token-level and time-specific fixed effects. We add a linear time trend ( $DD_i \times t$ ) to control for any other possible confounders induced by varying token distributions.

**Table 3: Heterogeneous Effect by Varying Degrees of Decentralization**

|                            | Contribution<br>Quantity <sup>2</sup> | Contribution<br>Originality <sup>2</sup> |
|----------------------------|---------------------------------------|------------------------------------------|
|                            | (1)                                   | (2)                                      |
| PD                         | -11.38***<br>(2.08)                   | -14.25***<br>(1.50)                      |
| PD $\times$ DD             | -11.25***<br>(4.30)                   | -6.27***<br>(2.11)                       |
| Avg. Price                 | 0.002***<br>(0.001)                   | -0.002<br>(0.003)                        |
| Return                     | -0.450<br>(0.47)                      | -0.005<br>(0.007)                        |
| Avg.MarketCap <sup>1</sup> | -0.001***<br>(0.0001)                 | 0.0001<br>(0.0002)                       |
| Avg.Volume <sup>1</sup>    | -0.003***<br>(0.0007)                 | 0.0001<br>(0.0001)                       |
| Individual Fixed Effects   | Yes                                   | Yes                                      |
| Time Fixed Effects         | Yes                                   | Yes                                      |
| Linear Time Trends         | Yes                                   | Yes                                      |
| Observations               | 6601                                  | 3602                                     |
| Token Units                | 189                                   | 172                                      |
| Adjusted R-Squared         | 0.54                                  | 0.83                                     |
| VCE                        | Robust                                | Robust                                   |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . <sup>1</sup>The unit of market cap and volume is in million USD.

<sup>2</sup>Contribution quantity is at the weekly level; contribution originality is at the monthly level.

Table 3 columns (1) and (2) report our estimates on the heterogeneous treatment effect for contribution quantity and originality, respectively. The treatment and moderating effects in both columns are negative and statistically significant. For contribution quantity, on average, a

decreased level of decentralization (increased level of centralization) will amplify the negative impact by contributing 11.25 fewer commits further. In terms of contribution originality, the decreased degree of decentralization will also exacerbate the negative impact by further reducing the originality level by 6.27 on average. More alarmingly, we observe that among the top quantile ventures ( $DD=1$ ), 53% of them cease to release a major new version after a P&D, while that of the counterparts ( $DD=0$ ) is 29%. In other words, developers of highly centralized blockchain ventures have a 180% higher probability of discontinuing major contributions compared to those of highly decentralized blockchain ventures.

This happens not just because centralized ownership can impede innovation (Hamel, 2011) but more importantly, it departs from the original design of tokens, which is to reward honesty and decentralize ownership (Nakamoto, 2008). An improper allocation of tokens may induce self-interest behaviors and distrust among participants and ultimately destroy the common welfare of all users, leading to the “Tragedy of the Commons”. Developers in a blockchain venture with a low degree of decentralization would have less control over the ongoing project since they hold a relatively smaller number of tokens. As a result, when their incentives, tokens, are distorted by P&D, they are more likely to cast doubt on the future of the venture and scale down or even discontinue their contributions to the project.

## **5. ROBUSTNESS CHECKS**

We conduct several robustness checks in this section.

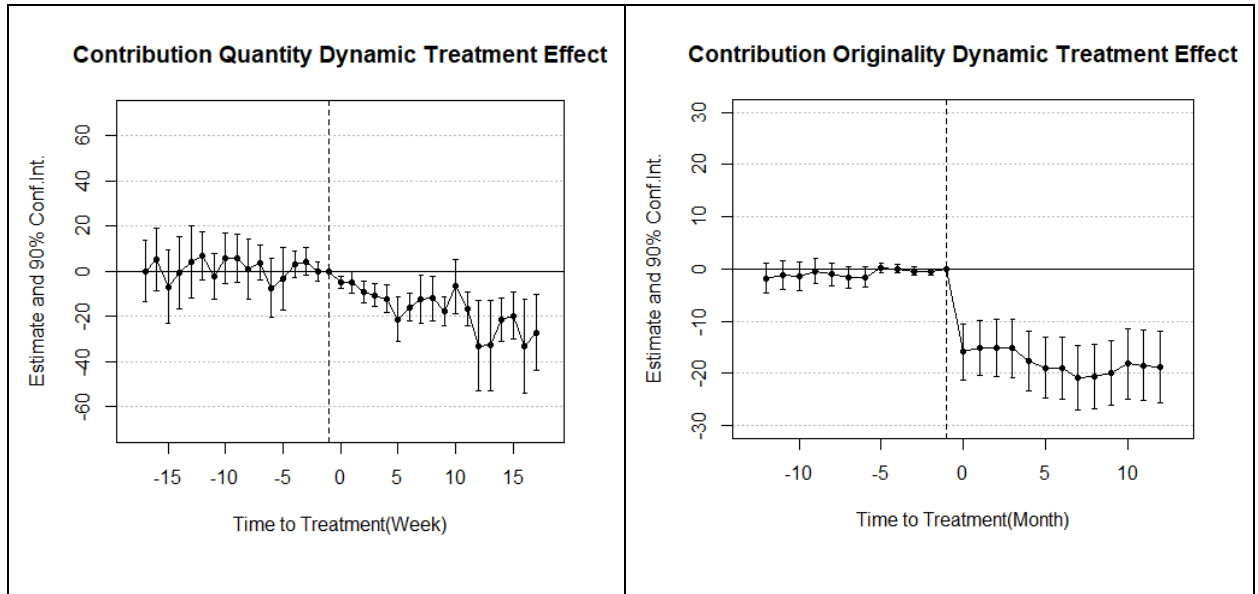
### 5.1. Dynamic Treatment Effects

It is common that the impact of treatment may not be homogenous over time. In our context, it is possible that the effect of P&D may vary at different times. We use Equation (3) to account for such dynamic treatment effects.

$$Y_{it} = \sum_{k=T_0}^{-2} \delta_k PD_{it}^k + \sum_{k=0}^{T_1} \delta_k PD_{it}^k + x'_{it}\beta + \alpha_i + \xi_t + \varepsilon_{it} \quad (3)$$

$PD_{it}^k$  is a dummy variable, where  $k$  indicates the relative time to token  $i$ 's treated period.

$PD_{it}^k$  is 1 for token  $i$  in time  $t$  if the relative time period is  $k$ , otherwise 0. For contribution quantity,  $T_0 = -17$  weeks, and  $T_1 = 17$  weeks. For contribution originality,  $T_0 = -12$  months, and  $T_1 = 12$  months. We dropped  $PD_{it}^{-1}$  to avoid multi-collinearity, following previous studies (Borusyak et al., 2021; Sun & Abraham, 2021). Figure 3 plots the estimates of dynamic treatment effects for each relative time period (see Table D1 in Appendix D for more details).



**Figure 3: Dynamic Treatment Effect for Contribution Quantity & Originality**

For all the pre-treatment periods, the treatment effect is null and statistically insignificant, whereas for post-treatment periods, the treatment effects become statistically significant and

negative. The treatment effect on contribution quantity exhibits a negative impact that gradually declines over time (with a stronger negative impact in the later periods). The treatment effect for contribution originality, on the other hand, has a substantial negative effect from the beginning and remains so throughout the subsequent periods. Our estimates corroborate what we observe in reality. Close to one-third (34%) of the blockchain ventures cease releasing new versions<sup>9</sup> shortly after P&D and only commit to resolving issues such as bug fixes. This observation explains why the dynamic treatment effect exhibits a progressive decline in quantity but a sharper decrease in originality. It also reveals the flaw of using the number of commits as a quality measure, echoing Lopes et al. (2017). Eventually, developers even cease committing to minor issues. As a result, the magnitude of the treatment effect becomes larger in later periods for contribution quantity. On a side note, this analysis of the dynamic treatment effect lends further support to the pre-treatment parallel trend assumption.

## **5.2. Addressing Selection Bias**

To further validate that P&D organizers do not target tokens based on criteria such as market cap, transaction volume, weekly return or token price, we apply the Probit selection model to all of the 255 tokens experiencing P&D and the rest 2,309 tokens without P&D (corresponding to the first stage of the Heckman selection model). These variables are commonly used in the P&D literature on stocks. Table 4 shows that none of these variables are statistically significant, and their coefficients are all very close to 0. This indicates that P&D organizers do not select target tokens based on trading criteria.

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<sup>9</sup> Appendix B explains the difference between GitHub release and commit. In brief, release indexes new functions or major updates to a project, while commit indicates small changes such as bug fixes.



**Table 4: Probit Model Estimates**

|                            | Coef.                | p-value |
|----------------------------|----------------------|---------|
| Avg.Prices                 | -0.0003<br>(0.0002)  | 0.120   |
| Avg.Weekly Return          | -0.0006<br>(0.0009)  | 0.492   |
| Avg.MarketCap <sup>1</sup> | 0.00003<br>(0.00002) | 0.155   |
| Avg.Volumes <sup>1</sup>   | 0.00001<br>(0.00007) | 0.843   |
| Target Token Units         |                      | 255     |
| Non-target Token Units     |                      | 2,309   |

<sup>1</sup>The unit of market cap and volume is in million USD.

Thus, we conclude that the selection of P&D tokens is rather random, regardless of their price, market performance, or return rate.

### 5.3. DID Specification with a Linear Time Trend

Besides the IFect model, we apply the conventional DID estimation here as a robustness check with Equation (4).

$$Y_{it} = \delta_{it}PD_{it} + x'_{it}\beta + \alpha_i + \xi_t + \gamma_i \times t + \varepsilon_{it} \quad (4)$$

The notations are the same as Equation (1). We include the individual-specific linear time trend  $\gamma_i \times t$  to explicitly control for any unobserved time-varying heterogeneity, as suggested by Mora and Reggio (2012). Likewise, we apply the model to the CEM-matched samples. The results reported in Table E1 in Appendix E are consistent with those of the IFect model.

### 5.4. Zero-Inflated Poisson Regression

The contribution quantity can be considered a count variable for which a zero-inflated Poisson model can be a viable alternative model. As demonstrated in Table E2 in Appendix E, both the main effects and the interaction term are consistent with the main model.

## 5.5. Removing Tokens with a High Turnover Rate

One concern with our token distribution data is that we are unable to obtain the complete historical data on token holder statistics. There might be some tokens whose ownership changes drastically after a P&D and this ownership change may not be captured by our token distribution data. We maintain that this may not pose a severe issue since the majority of our tokens exhibit a relatively small shift in holder statistics. Nevertheless, to address such concerns, we calculate the share turnover rate for all 255 P&D-treated tokens and remove those tokens with a high turnover rate in our analysis. Our share turnover rate is calculated using Equation (5) following Chen (2022).

$$\text{Share Turnover Rate}(\%) = \frac{24 \text{ Hour Trading Volume on event date}}{\text{Total Supply in the Market on event date}} \quad (5)$$

**Table 5: Heterogeneous Effects Summary without High Turnover Rate Token**

|                            | Contribution Quantity <sup>2</sup> |                       | Contribution Originality <sup>2</sup> |                      |
|----------------------------|------------------------------------|-----------------------|---------------------------------------|----------------------|
|                            | 95 <sup>th</sup> (1)               | 90 <sup>th</sup> (2)  | 95 <sup>th</sup> (3)                  | 90 <sup>th</sup> (4) |
| PD                         | -11.79***<br>(2.11)                | -10.17***<br>(2.12)   | -14.53***<br>(1.51)                   | -15.37***<br>(1.57)  |
| PD × DD                    | -12.49***<br>(4.45)                | -14.56***<br>(4.54)   | -5.65**<br>(2.17)                     | -3.90*<br>(2.14)     |
| Avg.Price                  | 0.002***<br>(0.001)                | 0.002***<br>(0.001)   | -0.002<br>(0.003)                     | -0.002<br>(0.003)    |
| Return                     | -0.45<br>(0.47)                    | -0.50<br>(0.47)       | -0.005<br>(0.007)                     | -0.005<br>(0.007)    |
| Avg.MarketCap <sup>1</sup> | -0.001***<br>(0.0001)              | -0.001***<br>(0.0001) | 0.0001<br>(0.0002)                    | 0.0001<br>(0.0002)   |
| Avg.Volume <sup>1</sup>    | 0.003***<br>(0.001)                | 0.003***<br>(0.001)   | 0.0001<br>(0.0001)                    | 0.0001<br>(0.0001)   |
| Individual Fixed Effects   | Yes                                | Yes                   | Yes                                   | Yes                  |
| Time Fixed Effects         | Yes                                | Yes                   | Yes                                   | Yes                  |
| Linear Time Trends         | Yes                                | Yes                   | Yes                                   | Yes                  |
| Observations               | 6,465                              | 6,363                 | 3528                                  | 3474                 |
| Token Units                | 185                                | 182                   | 168                                   | 165                  |
| Adjusted R-Squared         | 0.54                               | 0.54                  | 0.83                                  | 0.83                 |
| VCE                        | Robust                             | Robust                | Robust                                | Robust               |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . <sup>1</sup>The unit of market cap and volume is in million USD.

<sup>2</sup>Contribution quantity is at the weekly level; contribution originality is at the monthly level.

Figure F4 in Appendix F depicts the cumulative histogram of share turnover rate for all 255 treated tokens. We notice that 95% of the tokens have a rate of less than 18% during the P&D event, and 90% of tokens have a rate less than 13%. Hence, our stratification approach to identify the top quartile and bottom quartile is unlikely to change.

To further ensure the validity of our results, we remove tokens with a turnover rate in the top 5<sup>th</sup> percentile and top 10<sup>th</sup> percentile, respectively, from our data. We then re-conduct our analysis for RQ2 (pertaining to the heterogenous effect of decentralization). The results are reported in Table 5. As shown in columns (1) and (2), the results for contribution quantity maintain the same sign and similar magnitude. Likewise, the results for contribution originality, as shown in columns (3) and (4), are also consistent with our main results.

## **6. CONCLUSION AND DISCUSSION**

This study investigates the effectiveness of tokens as incentives in motivating developers of blockchain ventures to contribute. We analyze data from four platforms (PumpOlymp, CoinMarketCap, CoinGecko, and GitHub) and propose two novel metrics to measure developer contributions, focusing on both quantity and originality. The originality measure is constructed by leveraging granular code-level data based on the similarity of the code with all the code files created before. Our findings demonstrate that although tokens can provide a unique incentive structure for participants in blockchain ventures, they can be adversely affected by external pump-and-dump events, leading to a reduction in the quantity and originality of developer contributions. This can result in significant damage to the development and reputation of the blockchain venture. Our study underscores the importance of maintaining a reasonable and stable token value to incentivize developers effectively. Additionally, we examine the moderating role of

decentralization. Our empirical evidence suggests that a low degree of decentralization may exacerbate the negative impact of P&D, further demotivating developers to contribute.

To practitioners and investors, we provide a new strategy to evaluate blockchain ventures and cryptocurrencies that involves a thorough analysis of their technical details. We encourage investors to delve deep into the genuine technology breakthrough achieved by these blockchain ventures before making investment decisions rather than solely relying on the hype surrounding cryptocurrencies. To blockchain ventures, we provide insight into designing tokens that not only serve as short-term rewards but also align with the long-term goals and values of the venture. These tokens should be designed and distributed with constraints that ensure their sustainability and viability over the long run rather than just focusing on short-term gains. For instance, setting limits on the number of tokens that can be issued or held by specific individuals to prevent concentration of power; implementing mechanisms that incentivize long-term holding and discourage short-term sell-offs, such as staking or lock-up periods; enforcing transparency in token distribution and allocation to promote equity and accountability; and providing a clear utility for the token beyond just its value. To ICO founders, we recommend being attentive to token distribution among stakeholders when raising capital in ICOs. It is crucial for blockchain ventures to avoid leaving a large portion of tokens solely to a few founders to promote confidence and trust among developers and other stakeholders. To policymakers, we suggest that regulators work with industry stakeholders to develop proper standards and regulations for cryptocurrency exchanges and trading activities to prevent price manipulation and build trust in the blockchain ecosystem. To blockchain developers, we would like to share the comment from Vitalik, the creator of Ethereum: “Now that the awful opportunistic money people have been washed out, the chain has a bright future.”

There are several limitations in our study. First, we aggregate developers' contributions at the project level rather than at the individual developer level, which may not fully capture the nuanced impact of individual developers based on their expertise or tenure within the venture. The impact on developers who are part of the founding team may be greater than that on developers who haphazardly joined the project because the former are more dedicated to the venture. The influence on developers with higher expertise may be stronger than that on newer developers because they are more invested in their reputation. A potential future direction is to tag each commitment and version update to individual developers and investigate how the treatment effect may vary based on individual developer characteristics. Another limitation of our study is that we are unable to obtain the exact number of tokens rewarded to each developer. As such, it is difficult to capture the potential impact that may be influenced by the varying amounts of reward tokens received by each developer. Future studies can explore the token development pool for each blockchain venture to investigate the proper number of tokens to be rewarded to developers based on their contributions. Meanwhile, we are unable to obtain the identity of P&D organizers, and there could be some insiders involved in the P&D groups. In the future, researchers can delve into the impact of insider trading on developers' contributions. Lastly, we hope our research inspires future research efforts to create a more transparent, fair, and trustworthy blockchain environment. We conclude with the statement by Michael Casey, the founder of CoinDesk.

*“The good news is that the glare of public opinion will eventually dissipate and that as the spotlight diminishes, real developers will find themselves in a healthier environment within which to do the work needed to unlock this technology’s potential.”*

Michael J. Casey

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## ONLINE APPENDIX

### Appendix A: Cryptocurrency P&D Scheme

In the cryptocurrency market, the P&D scheme is a type of market manipulation that involves artificially inflating the price of a cryptocurrency and quickly selling off for a profit. Normally, P&D organizers utilize encrypted message apps such as Telegram to gather participants and post P&D signals. The signal information, including details such as target token symbol, exchange platform, and target return, is typically not released until the scheduled time (See Figure A1) shows an example of a signal announcement on Telegram). Participants have the option to pay an extra membership fee to gain access to the VIP channel, which provides early access to the target information before regular members (Figure A1) shows an instance of 5,336 VIP members in the VIP channel and 14,254 common members in the regular channel).

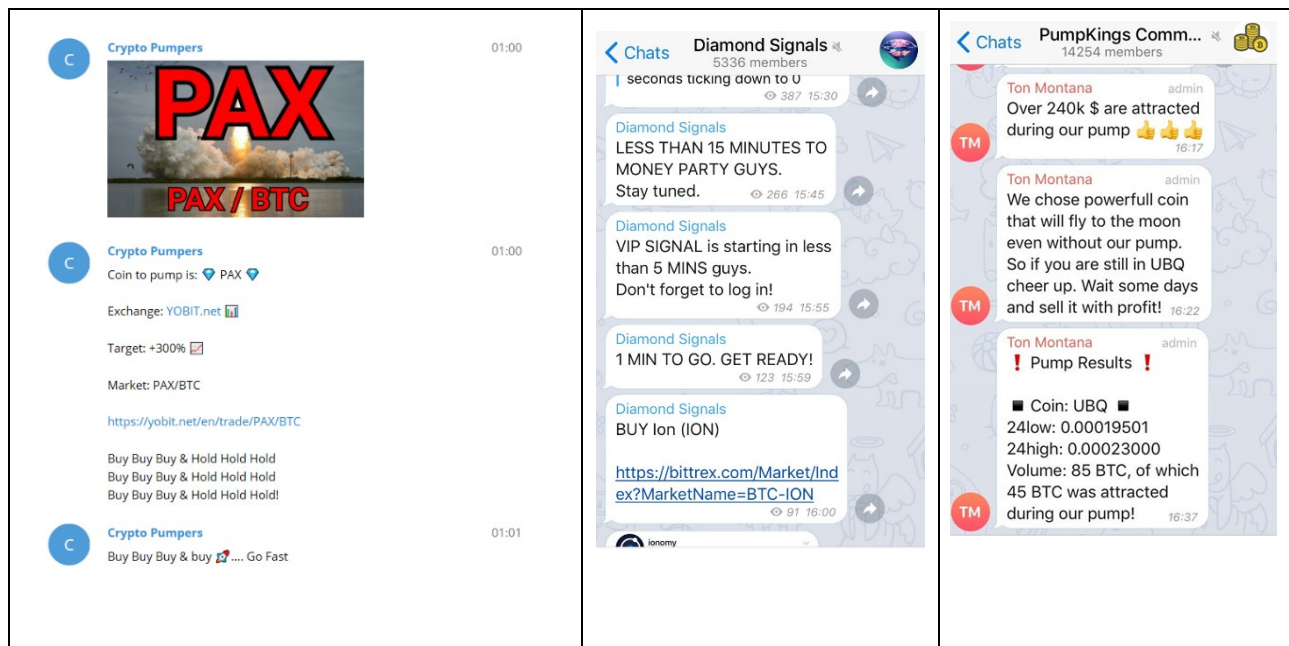


Figure A1: P&D Signal, VIP vs. Regular Channel on Telegram

Unlike P&D in the stock market, cryptocurrency P&D often focuses on trade-based manipulation by increasing transaction volume through inviting many participants to pump the

token price (Tsuchiya, 2021). In the pump phase, organized P&D groups use social media platforms such as Reddit and Twitter to advertise their information to attract more investors.



Figure A2: Advocate of Dogecoin on Reddit



Figure A3: YoBit Random Select P&D Coin

Figure A2 shows an example of P&D groups promoting Dogecoin on Reddit, which resulted in 56.7k likes and 4.2k comments. The pump phase is followed by the dump phase, during which the P&D groups sell off their shares, leading panicked investors to sell as well, causing a dramatic price decrease. The selection of tokens is rather random, as illustrated in Figure A3, which shows an example of random token selection at YoBit<sup>10</sup>. Token promotions tend to use false news or rumors to promote coins, rather than anything related to the quality of a token (Hamrick et al., 2018). Take Dogecoin as an example. In early 2021, with Elon Musk's and WallStreetBets' direct

<sup>10</sup> YoBit is one of the largest cryptocurrency exchange platforms and has one of the largest communities on Telegram with over 80,000 people.

advocacy, Dogecoin’s price surged more than 150% (Ostroff & Vigna, 2021). Similarly, meme coins such as SHIB also shared the hype. Nevertheless, the underlying value of these meme coins is debatable. In an interview with Fortune, the CEO of Block Journal, David Hsiao, and the head of research at the cryptocurrency exchange AAX, Ben Caselin, stated that the hype surrounding meme coins is a social phenomenon and would not be sustainable. The selection of coins is based on memes rather than technology or use cases like Ethereum (McGregor & Lau, 2021).

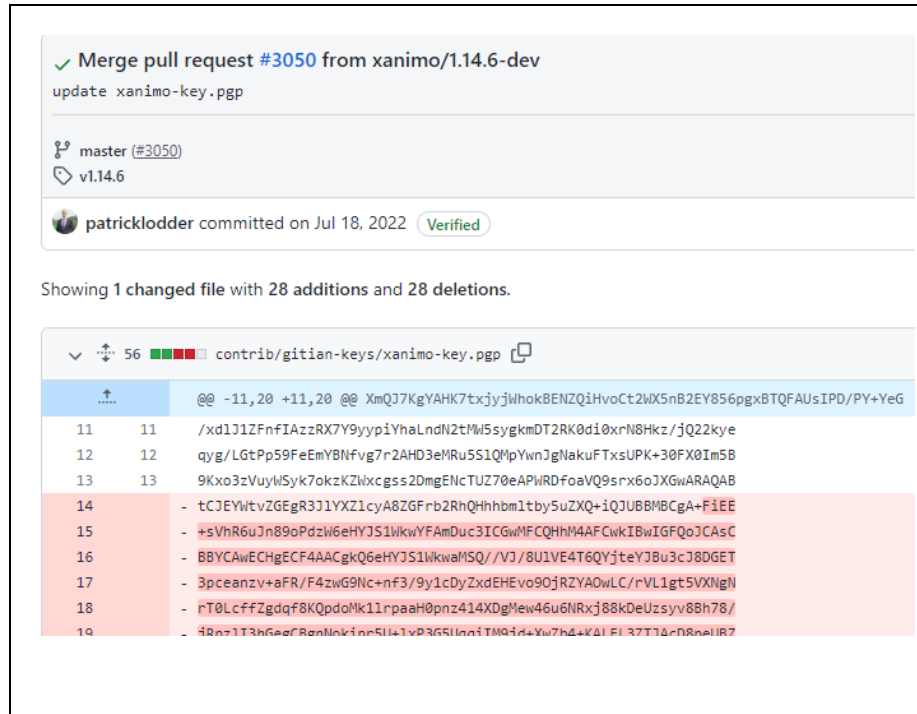
Once a P&D event is over, the organizers will post the profit information on the Telegram channel (Figure A1). Typically, a P&D event lasts fewer than three minutes, and more than 90% of them succeed with positive returns.

### **Appendix B: Commit and Release on GitHub**

Our two contribution measures, quantity and originality, are derived from two features tracked on GitHub: commit and release, respectively.

A commit on GitHub refers to a specific change made to a file or a set of files. These changes are normally associated with fixing “bugs”, addressing “issues” or resolving “defects” in the codebase that require more immediate attention. Usually, they are tagged with the specific release of the codebase (Figure B1 presents an example of Dogecoin’s commit tagged with release version 1.14.6). In general, a commit is a snapshot of the source code. It captures a specific moment in time and may not fully represent the overall progress, especially in terms of new features and breaking changes, making it an incomplete measure of developers’ contributions. The frequency of commits can reflect the current activity level of a project and its contributors, as it often reflects their ongoing efforts to address issues that demand prompt resolution, but it does not provide a comprehensive view of code quality or originality. In this paper, we follow the computer science

literature and use commits as a measure of quantity (Munaiah et al., 2017) rather than originality or quality.



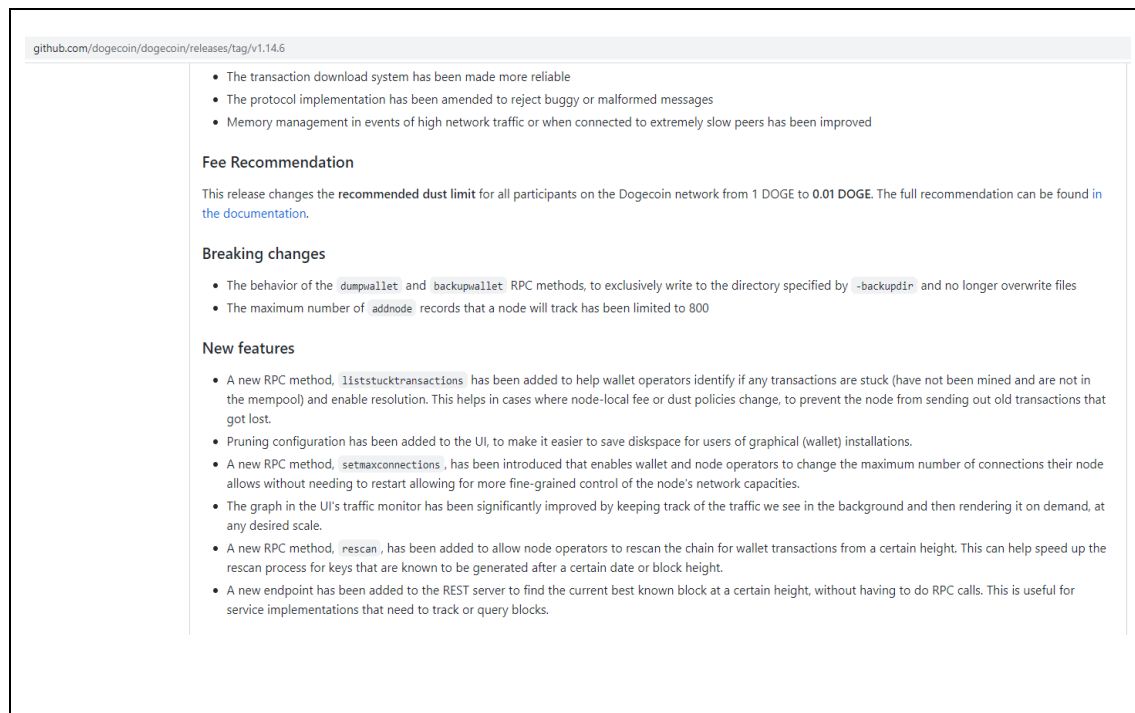
**Figure B1: Example of Dogecoin's Commit Tagged with Version 1.14.6**

A release is a way to package a specific version of a project's code for distribution, which typically contains the most recent stable and complete source code for a project. This is how we retrieved the source code for each version update and the original source code. A release can contain one or more commits, but more importantly, a release is a higher-level concept that includes more than just commits. It can include breaking changes and new features (Figure B2 presents an example of Dogecoin's release for version 1.14.6 with new features).

As an analogy, Microsoft Windows 10 and Windows 11 are both new releases of a software system that bring new features and major changes; but for each version of the system, there are update patches made to that system version to resolve issues and bugs. Likewise, releases on GitHub are major updates to a software project, while commits are changes made to specific files

within that version that typically are related to small changes such as a bug fix. As such, a new release can be a good indicator of new functions or features added to a project.

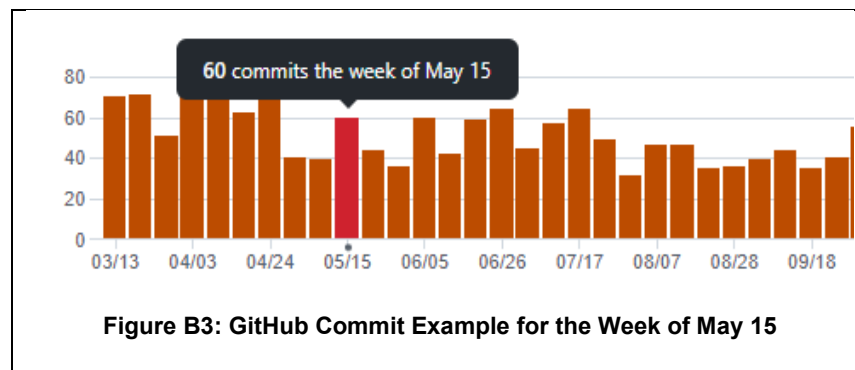
Moreover, a release includes a complete set of code files at the *file-level*, while a commit is a snapshot of a specific change made to a file or set of files at a particular point in time. Previous literature suggests that using commitment tags such as “bug(s)”, and “defects” to determine the quality of the project may overlook *file-level* duplication issues (Lopes et al., 2017), further supporting the use of source code files from each release for assessing code originality. Thus, we use new release version updates to construct the code originality measure.



**Figure B2: Example of Dogecoin's Release Version 1.14.6**

We follow the weekly aggregate level used by GitHub (Figure B3) and use the number of commits in a week to capture contribution quantity, which is a direct and immediate indicator of the current activity level of developers. We use the source code originality score in a month to represent a more comprehensive evaluation of contribution quality for the following reasons.

Major version updates are not as frequent as commits since the latter often address issues that require more immediate attention, while major version updates in each release are often related to new features, which typically require significant development time and effort. The average number of commits is 27.86 per week, while the average version release is 1.09 per month. The average time spent between two subsequent version releases is 27.5 days, which is around 1 month. Therefore, a weekly level is more suitable for the commit-based contribution quantity measure, while a monthly level is better for the originality measure.



## Appendix C: Additional Tables

### 1. Descriptive Statistics of the PumpOlymp Data

Table C1 provides the summary statistics for all the 1,339 P&D events across 403 cryptocurrencies tracked by PumpOlymp from June 2018 to July 2021.

**Table C1: PumpOlymp Event Table**

| Variables              | Description                                                    | Mean  | Std.Dev. | Min    | Max   |
|------------------------|----------------------------------------------------------------|-------|----------|--------|-------|
| Event Duration (s)     | Duration in seconds of each P&D event                          | 139.1 | 235.3    | 0      | 3,406 |
| Number of Events       | Statistics on the number of P&D events for each cryptocurrency | 3.3   | 4.0      | 1      | 24    |
| Theoretical Profit (%) | The theoretical profit achievable in a P&D event in percentage | 26.31 | 69.82    | -64.10 | 864.8 |
| Volume                 | The transaction volume of a P&D event                          | 65.94 | 291.2    | 0      | 6,683 |



## 2. Descriptive Statistics of the CoinGecko Data

Table C2 presents the summary statistics for all the 5,990 cryptocurrencies tracked by CoinGecko from April 2013 to December 2020, aggregated at a weekly level.

**Table C2: CoinGecko Data Summary**

| Variables                | Description                                          | Mean  | Std.Dev. | Min | Max       |
|--------------------------|------------------------------------------------------|-------|----------|-----|-----------|
| Market Cap (Million USD) | Weekly average market capitalization in millions     | 107.7 | 3261.3   | 0   | 369,999   |
| Volume (Million USD)     | Weekly average total transaction volumes in millions | 28.21 | 4,443.5  | 0   | 3,009,000 |
| Price (USD)              | Weekly average token price                           | 69.65 | 980.9    | 0   | 4,9855.7  |
| Weekly Commit            | The number of developers' commits per week           | 4.83  | 38.84    | 0   | 2,557     |

## 3. Statistics on Languages Used by Different Blockchain Ventures

Table C3 reports the distribution of different programming languages used by blockchain ventures for the 2,351 projects listed on GitHub. The top eight languages we focus on account for more than 75% of the projects. In total, our codebase data contains more than 1.27 billion code lines and 18,010 code files.

**Table C3: Distribution of Different Programming Languages**

| Programming Language | Number of Projects (Ventures) | Estimate Lines of code (Million) |
|----------------------|-------------------------------|----------------------------------|
| C++                  | 892                           | 1,060.5                          |
| JavaScript           | 430                           | 15.22                            |
| Solidity             | 145                           | 5.39                             |
| Go                   | 144                           | 64.18                            |
| TypeScript           | 130                           | 24.15                            |
| C                    | 67                            | 88.09                            |
| Python               | 62                            | 6.86                             |
| Java                 | 47                            | 10.52                            |
| Other 41 Languages   | 572                           | --                               |

Note: some ventures use more than one programming language.

## 4. Descriptive Statistics of the PumpOlymp Data

Table C4 tabulates the summary of holder statistics for all the 3,555 tokens whose holder information is tracked by CoinMarketCap.

**Table C4: Summary of Token Holder Statistics**

| Variables           | Description                                  | Mean  | Std.Dev. | Min | Max |
|---------------------|----------------------------------------------|-------|----------|-----|-----|
| Top 10 holders (%)  | Percentage of tokens held by top 10 holders  | 79.66 | 19.94    | 0   | 100 |
| Top 20 holders (%)  | Percentage of tokens held by top 20 holders  | 84.78 | 17.41    | 0   | 100 |
| Top 50 holders (%)  | Percentage of tokens held by top 50 holders  | 89.96 | 13.98    | 0   | 100 |
| Top 100 holders (%) | Percentage of tokens held by top 100 holders | 92.93 | 11.79    | 0   | 100 |

## 5. Covariate Balance After CEM

Table C5 and Table C6 present the covariate balance results for contribution quantity and originality using CEM, respectively. On average, the covariate balance was improved by more than 90% after matching.

**Table C5: Covariate Balance Before and After Matching for Contribution Quantity**

| Variables         | Before Matching<br>(Tr = 148, Con = 900) |                   | After Matching<br>(Tr = 123, Con = 260) |                   | Percent<br>Improvement |
|-------------------|------------------------------------------|-------------------|-----------------------------------------|-------------------|------------------------|
|                   | Mean<br>(Treated)                        | Mean<br>(Control) | Mean<br>(Treated)                       | Mean<br>(Control) |                        |
| Avg.logMarketCap  | 16.99                                    | 13.83             | 16.80                                   | 16.59             | 93.43                  |
| Avg.logVolume     | 13.93                                    | 9.17              | 13.70                                   | 13.25             | 90.51                  |
| Avg.Weekly Return | -0.0035                                  | 4.72              | 0.0058                                  | 0.0038            | 99.95                  |
| Avg.Volatility    | 0.30                                     | 0.59              | 0.283                                   | 0.299             | 94.53                  |

**Table C6: Covariate Balance Before and After Matching for Contribution Originality**

| Variables         | Before Matching<br>(Tr = 104, Con = 864) |                   | After Matching<br>(Tr = 100, Con = 244) |                   | Percent<br>Improvement |
|-------------------|------------------------------------------|-------------------|-----------------------------------------|-------------------|------------------------|
|                   | Mean<br>(Treated)                        | Mean<br>(Control) | Mean<br>(Treated)                       | Mean<br>(Control) |                        |
| Avg.logMarketCap  | 17.01                                    | 13.48             | 16.84                                   | 16.68             | 95.60                  |
| Avg.logVolume     | 13.68                                    | 8.25              | 13.44                                   | 13.35             | 98.22                  |
| Avg.Weekly Return | 0.00073                                  | 6.39              | 0.0011                                  | 0.0046            | 99.94                  |
| Avg.Volatility    | 0.426                                    | 1.15              | 0.431                                   | 0.489             | 92.09                  |

## 6. Change of the Dependent Variables

Table C7 reports the change in the dependent variables upon a P&D event for the CEM-matched data. Contribution quantity decreases by around 21.58 on average, while contribution originality decreases by about 14.18 on average.

**Table C7: Dependent Variable Statistics Before & After P&D Event**

| Variables           | Before P&D Event  |                   | After P&D Event   |                   |
|---------------------|-------------------|-------------------|-------------------|-------------------|
|                     | Mean<br>(Treated) | Mean<br>(Control) | Mean<br>(Treated) | Mean<br>(Control) |
| Weekly Commit       | 38.91             | 18.22             | 30.23             | 31.12             |
| Monthly Originality | 74.70             | 73.06             | 57.71             | 70.25             |

## 7. Summary Statistics for the Matched Samples

**Table C8: Summary Statistics for Contribution Quantity Matched Tokens**

| Variables                            | Description                                              | Mean   | Std.Dev. | Min   | Max     |
|--------------------------------------|----------------------------------------------------------|--------|----------|-------|---------|
| Weekly Avg. Market Cap (Million USD) | Weekly average market capitalization in millions USD     | 689.67 | 10,532   | 0     | 335,707 |
| Weekly Avg. Volume (Million USD)     | Weekly average total transaction volumes in millions USD | 116.65 | 1,590    | 0     | 48,809  |
| Weekly Avg. Price (USD)              | Weekly average token price                               | 84.74  | 1,166    | 0     | 24,477  |
| Weekly Return                        | Weekly return of each token                              | 0.02   | 0.82     | -0.99 | 52.8    |
| Weekly Commit                        | The number of developers' commits per week               | 27.86  | 96.37    | 0     | 1907    |

**Table C9: Summary Statistics for Contribution Originality Matched Tokens**

| Variables                             | Description                                               | Mean   | Std. Dev. | Min | Max     |
|---------------------------------------|-----------------------------------------------------------|--------|-----------|-----|---------|
| Monthly Avg. Market Cap (Million USD) | Monthly average market capitalization in millions USD     | 650.83 | 9,706     | 0   | 355,207 |
| Monthly Avg. Volume (Million USD)     | Monthly average total transaction volumes in millions USD | 121.29 | 1,585     | 0   | 52,161  |
| Monthly Avg. Price (USD)              | Monthly average token price                               | 29.40  | 521       | 0   | 19,132  |
| Monthly Return                        | Monthly return of each token                              | 0.101  | 1.86      | -1  | 85.13   |
| Monthly Originality                   | monthly contribution originality                          | 69.81  | 24.61     | 0   | 100     |

Table C8 and C9 report the summary statistics on the dependent variable and the main control variables for the two CEM-matched samples for contribution quantity (weekly) and originality (monthly), respectively.

## Appendix D: Dynamic Treatment Effect Results

Table D1 reports the dynamic treatment effect estimates for each time period (see Section 5.1).

**Table D1: Treatment Estimate Relative to Treatment Introduction**

| Contribution Quantity     |                   | Contribution Originality   |                   |
|---------------------------|-------------------|----------------------------|-------------------|
| Time to Treatment         | Estimate          | Time to Treatment          | Estimate          |
| 17 Weeks Before Treatment | 1.17 (7.98)       | --                         | --                |
| 16 Weeks Before Treatment | 6.19 (8.32)       | --                         | --                |
| 15 Weeks Before Treatment | -5.96 (9.63)      | --                         | --                |
| 14 Weeks Before Treatment | 0.048 (9.46)      | --                         | --                |
| 13 Weeks Before Treatment | 4.99 (9.52)       | --                         | --                |
| 12 Weeks Before Treatment | 7.60 (6.48)       | 12 Months Before Treatment | -1.82 (1.74)      |
| 11 Weeks Before Treatment | -1.76 (5.95)      | 11 Months Before Treatment | -1.10 (1.69)      |
| 10 Weeks Before Treatment | 6.12 (6.64)       | 10 Months Before Treatment | -1.44 (1.63)      |
| 9 Weeks Before Treatment  | 5.88 (6.45)       | 9 Months Before Treatment  | -0.42 (1.47)      |
| 8 Weeks Before Treatment  | 1.25 (8.08)       | 8 Months Before Treatment  | -1.00 (1.34)      |
| 7 Weeks Before Treatment  | 4.01 (4.73)       | 7 Months Before Treatment  | -1.61 (1.24)      |
| 6 Weeks Before Treatment  | -7.16 (7.87)      | 6 Months Before Treatment  | -1.59 (1.18)      |
| 5 Weeks Before Treatment  | -3.30 (8.48)      | 5 Months Before Treatment  | 0.22 (0.52)       |
| 4 Weeks Before Treatment  | 3.39 (3.58)       | 4 Months Before Treatment  | 0.005 (0.50)      |
| 3 Weeks Before Treatment  | 4.46 (3.66)       | 3 Months Before Treatment  | -0.46 (0.49)      |
| 2 Weeks Before Treatment  | 0.017 (2.65)      | 2 Months Before Treatment  | -0.52 (0.37)      |
| 1 Week Before Treatment   | --                | 1 Month Before Treatment   | --                |
| Treatment Introduced      | -4.66*** (1.59)   | Treatment Introduced       | -15.87*** (3.23)  |
| 1 Week After Treatment    | -4.79* (2.86)     | 1 Month After Treatment    | -15.14 *** (3.20) |
| 2 Weeks After Treatment   | -9.07*** (2.87)   | 2 Months After Treatment   | -15.12*** (3.39)  |
| 3 Weeks After Treatment   | -10.58*** (3.02)  | 3 Months After Treatment   | -15.14*** (3.40)  |
| 4 Weeks After Treatment   | -12.08*** (3.62)  | 4 Months After Treatment   | -17.63*** (3.41)  |
| 5 Weeks After Treatment   | -21.10*** (6.04)  | 5 Months After Treatment   | -18.89*** (3.56)  |
| 6 Weeks After Treatment   | -15.74*** (3.77)  | 6 Months After Treatment   | -19.04*** (3.59)  |
| 7 Weeks After Treatment   | -12.23* (6.45)    | 7 Months After Treatment   | -20.86*** (3.71)  |
| 8 Weeks After Treatment   | -11.82** (5.88)   | 8 Months After Treatment   | -20.69*** (3.74)  |
| 9 Weeks After Treatment   | -17.29*** (3.73)  | 9 Months After Treatment   | -19.85*** (3.73)  |
| 10 Weeks After Treatment  | -6.44 (7.39)      | 10 Months After Treatment  | -18.12*** (4.11)  |
| 11 Weeks After Treatment  | -16.30*** (4.50)  | 11 Months After Treatment  | -18.46*** (4.13)  |
| 12 Weeks After Treatment  | -32.62*** (11.96) | 12 Months After Treatment  | -18.79*** (4.16)  |
| 13 Weeks After Treatment  | -32.34*** (11.85) | --                         | --                |
| 14 Weeks After Treatment  | -20.75*** (5.66)  | --                         | --                |
| 15 Weeks After Treatment  | -18.90*** (6.14)  | --                         | --                |
| 16 Weeks After Treatment  | -32.22*** (12.33) | --                         | --                |
| 17 Weeks After Treatment  | -26.19*** (10.19) | --                         | --                |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

## Appendix E: Linear Trend Model and ZIP Model

We present the results of two additional robustness checks in this Appendix. Table E1 reports the treatment effect estimates using the DID specification with a linear time trend. The results are consistent with the main results. Both contribution quantity and originality remain negative and statistically significant (see Section 5.3).

**Table E1: DID Specification Results Summary**

|                            | Contribution<br>Quantity <sup>2</sup> | Contribution<br>Originality <sup>2</sup> |
|----------------------------|---------------------------------------|------------------------------------------|
|                            | (1)                                   | (2)                                      |
| PD                         | -6.92**<br>(2.48)                     | -14.49***<br>(1.28)                      |
| Avg.Price                  | 0.0004<br>(0.0005)                    | -0.007**<br>(0.002)                      |
| Return                     | -0.55<br>(0.34)                       | -0.001<br>(0.02)                         |
| Avg.MarketCap <sup>1</sup> | 0.0004<br>(0.0002)                    | 0.0004**<br>(0.0001)                     |
| Avg.Volume <sup>1</sup>    | 0.001<br>(0.001)                      | -0.0001*<br>(0.0001)                     |
| Individual Fixed Effects   | Yes                                   | Yes                                      |
| Time Fixed Effects         | Yes                                   | Yes                                      |
| Linear Time Trends         | Yes                                   | Yes                                      |
| Observations               | 13385                                 | 7221                                     |
| Token Units                | 383                                   | 344                                      |
| Adjusted R-Squared         | 0.77                                  | 0.93                                     |
| VCE                        | Robust                                | Robust                                   |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ <sup>1</sup>The unit of market cap and volume is in million USD.

<sup>2</sup>Contribution quantity is at the weekly level; contribution originality is at the monthly level.

Table E2 reports the main effect (1) and the heterogeneity treatment effect (2) for contribution quantity using zero-inflated Poisson Regression (see Section 5.4). The results are consistent with the main analysis.

**Table E2: Zero-Inflated Poisson Regression Results**

| Contribution Quantity    |                     |                     |
|--------------------------|---------------------|---------------------|
|                          | (1)                 | (2)                 |
| PD                       | -0.32***<br>(0.075) | -0.35***<br>(0.063) |
| PD × DD                  |                     | -0.56***<br>(0.15)  |
| log.avg.Price            | 0.056<br>(0.13)     | 0.101<br>(0.088)    |
| Weekly Return            | -0.11*<br>(0.056)   | -0.053<br>(0.087)   |
| log.avg.MarketCap        | -0.0099<br>(0.012)  | -0.016<br>(0.016)   |
| log.avg.Volume           | -0.0012<br>(0.015)  | -0.016<br>(0.016)   |
| Individual Fixed Effects | Yes                 | Yes                 |
| Time Fixed Effects       | Yes                 | Yes                 |
| Linear Time Trends       | Yes                 | Yes                 |
| Observations             | 13,385              | 6,601               |
| Token Units              | 383                 | 189                 |
| VCE                      | Robust              | Robust              |

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

## Appendix F: Additional Figures

This Appendix presents four additional figures used in the paper. Figure F1 illustrates an example of a C++ source code file in GitHub. Figure F2 shows two additional placebo and equivalence tests we conducted for contribution originality and quantity under different time periods (3 months and 2 months before, respectively; see Section 4.1).

```

// Copyright (c) 2009-2010 Satoshi Nakamoto
// Distributed under the MIT/X11 software license, see the accompanying
// file license.txt or http://www.opensource.org/licenses/mit-license.php.
#include "headers.h"
#include "db.h"
#include "net.h"
#include "init.h"
#include "auxpow.h"
#include "611.h"
#include "cryptopp/sha.h"
#include <boost/filesystem.hpp>
#include <boost/filesystem/fstream.hpp>

using namespace std;
using namespace boost;

//
// Global state
//

CCriticalSection cs_setpwalletRegistered;
set<CWallet*> setpwalletRegistered;

CCriticalSection cs_main;

boost::mutex mut_newBlock;
boost::condition_variable cv_newBlock;

map<uint256, CTransaction> mapTransactions;
CCriticalSection cs_mapTransactions;
unsigned int nTransactionsUpdated = 0;
map<COutPoint, CInPoint> mapNextTx;

CMapBlockIndex mapBlockIndex;
uint256 hashGenesisBlock("0x00000000bd61ac4b6a92d497274090be8448277becb90c91e70d275243da1d2");
// CBigNum bnProofOfWorkLimit(~uint256(0) >> 32);
CBigNum bnProofOfWorkLimit(~uint256(0) >> 20);

```

Figure F1: C++ Source Code Sample

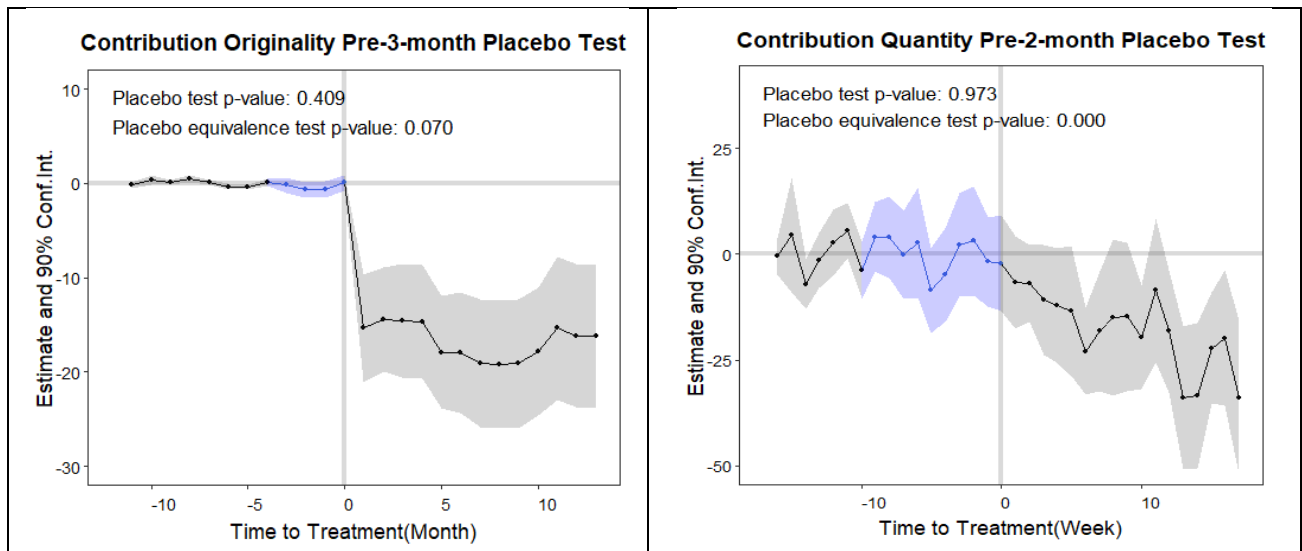


Figure F2: Placebo & Equivalence Test for Contribution Originality & Quantity

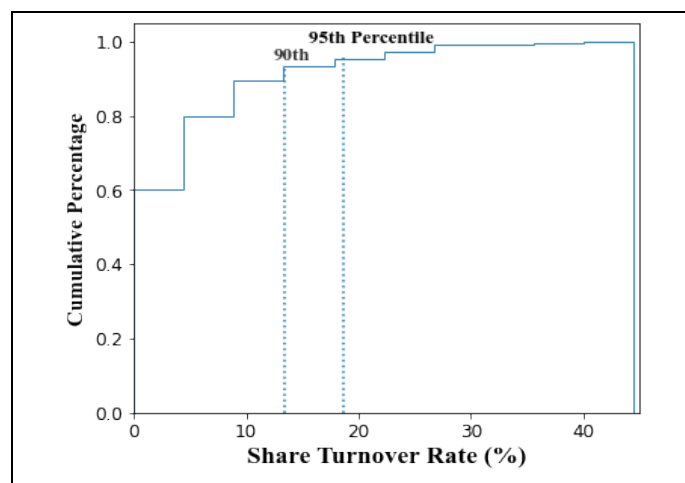
Figure F3 depicts CloakCoin's Market Price Chart from July 2014 to November 2015. The price peak corresponds to the P&D event in August 2014.

**CloakCoin Chart**



**Figure F3: CloakCoin Market Price Chart (Source: CoinMarketCap)**

Figure F4 shows the cumulative histogram of share turnover rate for all 255 treated tokens. Notes that 95% of the tokens have a share turnover rate of less than 18%, and 90% of tokens have a share turnover rate of less than 13% (see Section 5.5).



**Figure F4: Cumulative Histogram of Share Turnover Rate for Treated Tokens**